

 Scoreboard Internal SATA Hard Drives							
BRAND MODEL	Western Digital RaptorWD740ADFD	Samsung HD080HJ	Western Digital RaptorWD1500ADFD	Samsung HD160JJ	Western Digital Caviar RE16 WD1600YS	Maxtor DiamondMax 10	Western Digital Caviar RE16 WD2500YS
							
Price (Rs)	7,500	1,900	11,000	2,425	3,200	3,500	4,500
+	+ Ultimate performance	+ Lowest-priced	+ Ultimate performance	+ None in particular	+ Good performance	+ Value for money	+ Good performance
-	- Too expensive	- High cost per GB	- Too expensive	- Relatively slow	- Less value for money	- SATA/150	- Less value for money
Final Score (Out of 100)	61.62	61.44	62.71	64.11	67.84	72.83	69.79
Performance (Out of 70)	58.04	46.21	57.77	40.24	49.22	46.89	49.62
Cost/Capacity Index (Out of 30)	3.59	15.24	4.93	23.87	18.62	25.94	20.18
Features							
Formatted Capacity (GB)	74	80	150	160	160	250	250
Actual Capacity (GB)	69.24	74.53	139.73	149.05	153.38	233.76	233.76
Interface Type	SATA	SATA II	SATA	SATA II	SATA II	SATA	SATA II
Number of Platters	1	1	2	2	2	NA	3
Number of Recording Surfaces	2	2	4	4	4	NA	6
Rotational Speed (RPM)	10000	7200	10000	7200	7200	7200	7200
Buffer Memory (MB)	16	8	16	8	16	16	16
Performance							
Synthetic Tests							
HD Tach RW 3.0.1.0							
Average Read (MB/s)	75.4	51	78.1	52.7	55.9	56.8	55.6
Average Write (MB/s)	72.9	48.8	67.8	51	53.7	54.8	53.1
CPU Utilisation (%)	4	2	3	3	3	4	2
Random Access Time (ms)	8	13.2	7.9	13.5	12.9	14.8	12.9
Read Burst Speed (MB/s)	133.3	204.8	132.5	205.4	194.4	135.3	198.5
H2Bench							
Interface Speed (50%): Sequential Transfer (MB/s)	114	111.7	112.9	108.9	123.4	124.3	124
Repetitive Sequential Read "Core Test" (MB/s)	119.5	169.6	119.5	168.6	153.3	121.5	153.6
Random Access Read (ms)	8.06	13.16	8.02	13.43	12.94	13.37	13.39
Random Access Write (ms)	4.72	7.03	4.96	7.44	6.53	7.25	6.7
PCMark05							
HDD XP Startup (MB/s)	10.92	7.683	11.495	Fail	9.053	7.963	8.963
HDD Application Loading (MB/s)	9.058	6.02	9.49	Fail	7.622	6.415	7.518
HDD Virus Scan (MB/s)	96.787	77.447	95.903	Fail	92.835	87.904	91.424
SiSoft Sandra Pro Business 2007							
Drive Index (MB/s)	72	52	73	52	55	52	55
Sequential Read (MB/s)	83	59	85	61	62	62	62
Random Read (MB/s)	55	41	56	42	44	35	44
Sequential Write (MB/s)	81	59	82	50	62	63	63
Random Write (MB/s)	58	44	59	40	43	48	44
Access Time (ms)	6	8	6	8	7	13	7
Real-World Tests							
Assorted 4 GB File Write (sec)	64.6	83.1	64.5	80	78.1	75.4	68.3
Assorted 4 GB File Read (sec)	74.3	88.1	77.2	83.9	82.4	81.2	82.4
Assorted 4 GB Internal Copy (sec)	137.3	187	132.2	161.8	170.8	155.8	170
Sequential 4 GB File Write (sec)	56	75.6	56.2	80.3	68.1	66.5	68.1
Sequential 4 GB File Read (sec)	57.5	79.6	59.5	86.3	67.9	66.4	67.7
Sequential 4 GB Internal Copy (sec)	136.7	180	132.5	191	163	142	161.2
Photoshop CS2 Startup Time (sec)	11.7	9.8	12.2	12.1	10.5	10.6	11.4
Image Load (550 MB file) (sec)	39.6	39.5	39.3	43.3	42.1	42.9	45.8
Image Load (1 GB file) (sec)	68.5	94.5	95.8	97.6	84.1	91.8	84.8
FarCry Level Load Time (sec)	26.9	29.6	26	30	27.6	29.8	27.4
Price Per MB (Paise)	10.58	2.49	7.69	1.59	2.04	1.46	1.88